SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

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Intelligent Agents in a Smart Home Automation System:

Introduction:

A smart home automation system uses intelligent agents to control lighting, temperature, and security based on user behaviour and environmental conditions, it uses sensors to collect information and makes decisions to improve comfort, save energy, and ensure safety.

Types of intelligent agents:

1. Simple Reflex Agent:

A simple reflex agent makes decisions based on the current situation using condition-action rules.

Ex: Turns ON lights when motion is detected.

2. Model-Based Agent:

A model-Based Agent maintains information about the environment and uses past and current data to make decisions.

Ex: Remembers whether a room is occupied.

3. Goal-Based Agent:

A goal-based agent selects actions to achieve specific goals.

Ex: Advantage that helps achieve desired results.

Disadvantage of planning may take more time.

4. Utility-Based Agent:

A utility-based agent selects actions by balancing different factors, such as comfort, cost, and energy consumption.

Ex: Maintains a comfortable temperature while using reducing electricity usage.

5. Learning Agent:

A learning agent improves its performance by learning from past experiences and user preferences.

Ex: Learns the user's daily routine and adjusts lights automatically.

Conclusion:

A hybrid intelligent agent makes smart home automation more efficient, comfortable, secure, and personalized. It can respond quickly to environmental changes while learning user perferences over time.

Search Algorithms for a Robot in an Unknown Maze:

Introduction:

A robot placed in an unknown maze must find a path to the exit without prior knowledge of the layout. Uniformed search algorithms, such as uniform cost search (ucs) and Iterative deepening search (IDS), help the robot explore possible paths and reach the goal.

1. Uniform cost search (ucs):

Uniform cost search expands the position with the lowest total path cost first. It uses a priority queue to select the next possible to explore.

Working:

1. Start from the initial position
2. Identify the available paths
3. Explore the position with the lowest cost
4. Continue until the exit is found.

Advantages:
- Finds the least cost path when step costs are positive.
- Complete, under suitable positive-cost conditions.
- Useful when different paths have different costs.

Disadvantages:
- Requires more memory.
- Can take considerable time when the maze is large.

Time and space complexity: Both can be exponential in the worst case.

2. Iterative Deepening search (IDS):

Iterative deepening search performs depth-limited searches repeatedly, increasing the depth limit each time until the end exist is found.

Working: • Increase the depth limit by one
• Repeat until the exist is reached.

Advantages: • Uses less memory than UCS
• Complete when the branching factor is finite

Disadvantages: • Repeats some searches
• Doesn't not guarantee the least-cost path when movement costs differ.

Time complexity: $O(b^d)$

Space complexity: $O(bd)$

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